

Assignment 1

Econ 8007 Macroeconomic Theory I, Fall 2026

Monday, September 14, 2026

Part A: Work with Some Labor Market Data

Table 1 lists every series used in this assignment; all are free downloads from FRED (<https://fred.stlouisfed.org>), seasonally adjusted, monthly.

Symbol	Object	FRED ID	Sample
U_t	Unemployment level (thousands)	UNEMPLOY	1948:M1–
U_t^s	Unemployed less than 5 weeks (thousands)	UEMPLT5	1948:M1–
L_t	Civilian labor force (thousands)	CLF160V	1948:M1–
—	NBER recession indicator	USREC	monthly

Table 1: Data used in Parts A and C.

These definitions are helpful. We denote the unemployment rate by $u_t = U_t/L_t$ and the short-term share by $u_t^s = U_t^s/L_t$.

Questions

- A1. **Two-state hazards from duration stocks.** The lake model needs one separation rate s_t and one finding rate f_t . Following Shimer (2012), construct both from unemployment-duration stocks. Anyone unemployed at $t + 1$ either became unemployed during the month (and so has duration under five weeks) or was unemployed at t and did not find a job. In shares of the labor force,

$$u_{t+1} = u_{t+1}^s + (1 - f_t) u_t \implies f_t = 1 - \frac{u_{t+1} - u_{t+1}^s}{u_t}.$$

Given f_t , back out s_t from the lake-model law of motion $u_{t+1} = (1 - f_t)u_t + s_t(1 - u_t)$:

$$s_t = \frac{u_{t+1} - (1 - f_t) u_t}{1 - u_t}.$$

Plot s_t and f_t with NBER recessions shaded and report the sample means $(\bar{s}, \bar{f})$ —these are your calibration targets in Part C. Describe the cyclical behavior of each hazard in a few sentences: which one moves as a recession begins, which one governs the slow recoveries, and what has happened to the level of f_t since 2000? Finally, explain in one paragraph what this two-state method buries: which CPS transitions (through nonparticipation N) are being folded into s_t and f_t ?

- A2. **The flow steady state.** Compute the flow steady state $u_t^{ss} = s_t / (s_t + f_t)$ month by month and plot it against the actual unemployment rate u_t . Report the sample means of u_t and u_t^{ss} , the mean convergence factor $1 - \bar{s} - \bar{f}$, and the implied half-life $\ln 2 / |\ln(1 - \bar{s} - \bar{f})|$ in months. Explain in two or three sentences why u_t^{ss} tracks u_t so closely (the half-life is the answer), and point to the episodes where the gap is largest.

Part B: Theory

- B1. **McCall with separations.** Add exogenous separations to the McCall model: an employed worker loses her job with probability σ at the end of each period and returns to unemployment.

- (a) Write the Bellman equations for $W(w)$ and U , and derive the reservation wage equation

$$w^* - b = \frac{\beta}{1 - \beta(1 - \sigma)} \int_{w^*}^{\bar{w}} (w - w^*) dF(w).$$

- (b) Show that $\partial w^* / \partial \sigma < 0$ and interpret.
- (c) Let F be uniform on $[0, 1]$ and $\beta = 0.99$. For $\sigma \in \{0, 0.02, 0.05\}$ and $b = 0.3$, compute w^* , the job-finding probability $\lambda = 1 - F(w^*)$, and the steady-state unemployment rate $\sigma / (\sigma + \lambda)$. Then ask the model to deliver the finding rate you measured in A2 (say $\lambda = 0.45$): what value of b is required at $\sigma = 0.02$? Explain what this says about sequential search with one offer per period as a theory of the finding rate, and what the DMP model changes.

- B2. **The wage equation, in and out of steady state.** In the DMP model of the notes (discrete time, matches formed in t produce from $t + 1$, separations with probability σ at the end of the period):

- (a) Derive the steady-state surplus $S = (z - b) / [1 - \beta(1 - \sigma) + \beta\gamma\lambda_w(\theta)]$ and the wage equation $w = (1 - \gamma)b + \gamma(z + \kappa\theta)$, stating exactly where free entry and $\lambda_w = \theta\lambda_f$ are used.
- (b) Show that the wage equation holds period by period along any equilibrium path, $w_t = (1 - \gamma)b + \gamma(z + \kappa\theta_t)$, even when θ_t is not constant. Which terms cancel, and why?
- (c) Suppose instead that the wage is bargained once, at the time of hiring, and never renegotiated. Write the Nash problem for a match formed at t and explain (no need to solve) why the resulting wage depends on the expected path of θ from t onward.

Part C: Computation

Use $\beta = 0.996$ (so $r = 0.004$), $z = 1$, $b = 0.4$, $\eta = 0.5$ and $\gamma = \eta$ throughout, unless a question changes them. Tightness has no natural units in the data you constructed, so *normalize* the steady state to $\theta^* = 1$, as the textbook by Azzimonti et al. does.

- C1. **Calibrate to your hazards.** Set $\sigma = \bar{s}$ from A1, and choose χ and κ so that the steady state has $\lambda_w(\theta^*) = \bar{f}$ and $\theta^* = 1$. (Two targets, two parameters: the normalization makes $\chi = \bar{f}$ immediately, and the equilibrium condition then gives κ in closed form.) Report the parameter values and the steady state: $\lambda_w, \lambda_f, w^*, u^*, S$ and J , together with the implied mean durations of unemployment and of a vacancy. Compare u^* with the sample means of u_t^{ss} and u_t from A2 and explain any gaps (what does the two-state abstraction leave out?).
- C2. **Comparative statics.** At your calibration, re-solve for $z \in \{0.98, 1.02\}$, for $b = 0.7$, and for σ equal to the peak monthly separation rate in your A1 series. Report θ, λ_w, u and w in a table. Compute the elasticity $\varepsilon_{\theta z} = d \ln \theta / d \ln z$ numerically from the $\pm 2\%$ experiments. Plot u^* as a function of $b \in [0, 0.95]$, and find the value of b at which the (numerically computed) elasticity reaches 20. Comment: what has happened to the surplus $z - b$ at that point?
- C3. **The Beveridge loop.** Starting from your baseline steady state, simulate a permanent, unanticipated 2% increase in z at $t = 0$, reversed at $t = 12$, for $t = 0, \dots, 36$, using $u_{t+1} = (1 - \lambda_w(\theta_t))u_t + \sigma(1 - u_t)$ and $v_t = \theta_t u_t$. Plot the path in the (u, v) plane over the model's Beveridge curve $u = \sigma / (\sigma + \lambda_w(\theta))$. Report the impact response of v , its long-run response, and the number of months it takes u to close half the gap; state the direction of the loop. Connect to A1: at recession onsets your measured s_t increases while the model holds σ fixed. What feature of the entry into recessions will the model therefore miss, and which margin is forced to do all the work?